
Two Coding Agents Optimize GraphViz: A Controlled Comparison

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Abstract

Can a coding agent find and fix a real performance bottleneck on its own, with no hint about where the program spends its time? We gave two agents — Claude Code and Codex — the same untouched copy of GraphViz and the same instruction: make the dot layout program faster without changing what it draws. Each agent ran four times, eight runs in all, and no run could see the benchmark we would judge it on. Half the runs got an extra instruction to use the machine’s idle cores.

Every run found the same hot code. What differed was how far each went. All four Claude runs improved three of the five layout phases, making layout $1.39\times$ to $4.02\times$ faster across three graph shapes at two sizes. Of the four Codex runs, one improved a single phase ($1.02\times$ to $1.33\times$) and three shipped no code at all — each built something, measured it, found it slower or not worth keeping, and reverted. We report those as decisions rather than failures, and they left us with three binaries that were the unmodified program under another name, which is a useful check on our own noise estimates.

Two results surprised us. Repeating a prompt changed the outcome more than changing the agent did: Claude’s two identical runs differ by $1.39\times$ against $1.67\times$ on one graph, and Codex’s differ in whether anything shipped. And asking for concurrency is not the same as getting it — two Claude runs look equally parallel end to end, but re-running each on a single thread shows one owes almost nothing to its threads on the smaller graphs and a third of its result on the larger ones, while the other owes them most of its result on wide graphs and nothing at all on narrow ones. Whether a parallel binary is really a parallel result depends on the graph you hand it. How much a phase is worth depends on both the shape of the graph and its size: crossing minimization is 12.4% of layout time on one shape and 64.8% on another with the same number of nodes, and doubling the nodes cuts its share on every shape. Which phase deserves the effort is a property of the input, and it moves. We measured how far apart two runs of the *unmodified* program land, and report only differences larger than that.

1 Introduction

Coding agents are now routinely asked to modify real software. Most evaluations of that ability measure whether an agent can make a failing test pass [5] — a task with a clear oracle, where success is checkable by running the suite. Performance optimization has no such oracle. An agent asked to make a program faster must first decide *what* to make faster, and a plausible-looking change that helps nothing is indistinguishable from a real improvement unless someone measures carefully.

This paper asks a narrow version of that question. Given a mature C codebase, no hint about where time is spent, and an instruction only to make the program faster without changing its output, can a coding agent find a genuine bottleneck and fix it?

We use the dot layout engine from GraphViz [3]. It is real, widely deployed software rather than a benchmark written for the occasion; its layered pipeline [7, 4] has distinct phases whose costs can be measured separately, so a speedup can be attributed rather than merely observed; and its text output is deterministic, which makes “the output did not change” checkable rather than a judgment call.

What we did. Two agents — Claude Code and Codex — each ran four times, for eight runs in total. Each run received its own pristine checkout of GraphViz pinned to commit 23b06521 and a prompt naming no target function, file, or phase. Two runs per agent used a base prompt; the other two added a section asking the agent to treat concurrency as its primary strategy, because no run under the base prompt had introduced any. Every prompt forbade the shortcuts that would fake a win: caching layout results, special-casing inputs, and reducing the iteration counts that govern layout quality. No run had access to the evaluation harness, the graphs it uses, or any other run’s sandbox. We rebuilt all eight results ourselves and measured them against an untouched baseline, on three graph shapes at two node counts. This is an observational case study of eight runs: we report what each run did and what it bought, and we do not rank the agents or claim that either is generally better.

What we found. No run failed to locate expensive code. All eight reports name the same two regions — crossing minimization, and the network simplex serving ranking and coordinate assignment — and every binary that shipped produced byte-identical layout output.

They separate on breadth. All four Claude runs went into network simplex as well and improved three phases, measuring $1.39\times$ to $4.02\times$ end-to-end. Of the four Codex runs, one improved crossing minimization only ($1.02\times$ to $1.33\times$), and *three shipped no layout code at all*: each built a change, measured it against the workload the prompt described, found it slower or unprofitable, and reverted it. Those three binaries are the baseline recompiled, which we report as a deliberate outcome rather than a failure — and which gave us an unplanned null sample showing our per-phase noise floor is optimistic.

Repetition mattered more than we expected. Claude’s two base-prompt runs differ by $1.39\times$ against $1.67\times$ on one shape; Codex’s differ in whether anything shipped at all. A single run of an agent does not characterize it.

The parallelism prompt produced the subtlest result. Its two Claude runs land close together end-to-end, but re-measuring each at one thread shows they are not the same kind of result. One put its concurrency in crossing minimization and is rewarded on wide graphs at any size — threads account for $2.78\times$ of its result on the densest shape — and not at all on narrow ones. The other put its concurrency in network simplex behind large size thresholds: its threads are worth 1 to 16% at 5,000 nodes and 21 to 36% at 10,000, when the input finally reaches the size it was written for. Each design pays exactly where the input makes its phase expensive.

Which phase dominates is a property of the input, and two things move it. At identical node count, shape moves it hardest: crossing minimization ranges from 12.4% to 64.8% of layout time. Size moves it too, and always the same way — doubling to 10,000 nodes cuts that phase’s share on every shape and hands the difference to network simplex. Each combination sets a different ceiling, and on the sparsest shape every Claude run exceeds the ceiling available to any crossing-minimization optimization at all. The ceilings themselves fall with size, so the strategy of optimizing that one phase is worth least on the largest graphs.

Contributions.

- A controlled protocol for evaluating agent-driven optimization: balanced command ordering, which also yields an empirical noise floor from the baseline’s own spread across orderings, and a held-out evaluation harness.
- Phase-resolved measurements of eight agent runs across three graph shapes at two node counts, with per-phase attribution and the input-dependent ceiling that bounds it, plus repetitions that show how far two runs of one prompt can diverge.
- A thread-resolved decomposition of the runs that shipped concurrency, separating what their threads buy from what their serial changes buy — the measurement that decides whether a parallel-looking result is one.

- Evidence that a floor built from the baseline against itself understates cross-binary variation, from three shipped binaries that turned out to contain no change.
- A complete artifact: pinned source, every run’s diff, commit history and report, raw timing data, and the figure-generation scripts.

Section 2 covers background. Section 3 describes the protocol. Section 4 reports measurements. Sections 5–6 interpret them and state what we did not verify.

2 Background and Related Work

The dot layout pipeline. GraphViz’s dot engine draws directed graphs using the layered approach introduced by Sugiyama et al. [7] and refined by Gansner et al. [4]. It runs five phases in order, each depending on the last: *initialization* parses the file and builds the graph, *ranking* decides which layer each node belongs to, *crossing minimization* orders nodes within a layer to reduce edge crossings, *coordinate assignment* fixes their x positions, and *spline routing* draws the edges.

Ranking and coordinate assignment are both solved with network simplex. That matters later: one agent optimized the shared code and improved both. Crossing minimization is an iterative heuristic.

Agents modifying real code. Benchmarks such as SWE-bench [5] evaluate agents on resolving real repository issues, where a test suite provides the success signal. Performance work differs in that no test can tell you whether an agent found the *right* thing to optimize, and a change that is correct but useless passes every test. Prior work on learned compiler optimization [2] targets transformations chosen in advance; here the choice of target is the object of study.

3 Method

The design goal was that source code be the *only* thing differing between the binaries we compare. Everything below follows from that.

3.1 Sandboxes and build

Nine clones of GraphViz, all at commit 23b06521. One stays untouched as the baseline. The other eight are agent runs: two agents $\times$ two prompts $\times$ two repetitions, each on its own branch so the commit histories survive as artifacts.

The repetitions exist because a single run of an agent cannot distinguish what the agent tends to do from what it happened to do once. Every run started from the same commit with no knowledge of the others.

All nine were built with identical CMake flags (`-DCMAKE_BUILD_TYPE=Release`, Poppler and GVEDit disabled) and installed to separate prefixes. We checked that each dot loads its own build of the layout libraries rather than a shared one.

3.2 The prompts

Every run received one of two prompts. The base prompt stated the goal — make dot faster on large graphs without changing its output — and deliberately named no function, file, or phase. Finding the hot code was part of the task under study. It also:

- described the target workload shape (large layered graphs with heavy edge crossings) without supplying the evaluation graphs, so agents generated their own test inputs;
- required byte-identical `-Tdot` output and pointed at the 260 graphs in the repository’s own `tests/` directory;
- enumerated disallowed shortcuts: detecting or special-casing specific inputs, caching layout results across runs, reducing iteration counts or loosening convergence thresholds, hardcoding computed values, and modifying the test suite;

- stated that measurement would be independent, that identical builds vary by several percent on the target machine, and that a modest honest result or no result at all was an acceptable outcome.

The iteration-count prohibition matters most. Crossing minimization is iteration-bounded, so cutting its limit yields a large, real, reproducible speedup that is not an optimization. The prompt asked for that to be reported as a finding rather than shipped as a change.

The second prompt is the base prompt with one section added. It observes that dot is single-threaded while the machine’s other cores sit idle, asks the agent to treat concurrency as the primary strategy rather than an afterthought, suggests where independent work might be found (loops over ranks, components, or candidate orderings), and states that output must not depend on thread count or scheduling. It also asks for thread counts and per-thread-count speedups in the report, and for an explanation wherever the agent concluded something could not usefully be parallelized. Everything else — workload, constraints, disallowed shortcuts, deliverable — is word for word the base prompt.

We added it because no run under the base prompt introduced concurrency of any kind, in either agent. Whether that is what agents prefer or merely what they did here is exactly the sort of question a second prompt can separate.

Agents worked autonomously with no turn or time limit, committing incrementally and writing a report of what they profiled, changed, abandoned and verified. Neither had access to the evaluation harness, the benchmark graphs, or the other agent’s sandbox.

3.3 Evaluation graphs

Graph generation is deterministic. Nodes are split into equal layers and each points at fanout random nodes in the next layer, which is what creates the crossings layout has to resolve.

Three shapes are generated at each of two node counts, 5,000 and 10,000. The shape parameters — nodes per rank and fanout — are held fixed across sizes, so doubling the node count doubles the number of ranks and leaves rank width alone (Table 1):

Table 1: Evaluation graphs: three shapes at two node counts. **Ranks:** nodes $\div$ nodes-per-rank. **Edges:** (ranks $- 1$) $\times$ nodes-per-rank $\times$ fanout.

Shape	Nodes/rank	Fanout	5,000 nodes		10,000 nodes	
			Ranks	Edges	Ranks	Edges
sparse-deep	10	2	500	9,980	1,000	19,980
default	100	3	50	14,700	100	29,700
dense-shallow	200	4	25	19,200	50	39,200

These parameters keep every phase measurable. Wider ranks make crossing minimization so expensive that nothing else is visible next to it.

The generator does not pin ranks. Wrapping each layer in rank=same is legal DOT, but it supplies the answer the ranking phase exists to compute: on otherwise identical graphs that reduces ranking from 267 ms to 1 ms and removes the phase from measurement.

3.4 Measurement

All timings use `hyperfine 1.20.0` [6] with 10 warmup and 10 timed runs. All binaries for a graph are measured in one invocation, seconds apart under the same thermal conditions; separate invocations produce order-dependent bias large enough to swamp the effects we are looking for.

`hyperfine` runs all repetitions of one command before starting the next, so a fixed order always hands the same binary the tail of any drift. We rotate the starting position and pool: four orderings, 40 timed runs per binary per graph. With nine binaries a full rotation would take nine orderings; four is enough that no binary sits at the tail of drift in more than one of them, and it keeps a single invocation under an hour.

The rotation also supplies the noise floor. The baseline is measured four times, once per ordering, so the spread among its own means is what the protocol reports for a program compared against itself.

Binaries that use threads pick their own thread count, and we measure whatever they pick rather than imposing one: that is what a user running the binary would get. The two differ — one defaults to every online core, the other to the performance cores only — so Section 4.7 measures each of them again at one thread, which is what separates a threaded gain from a serial one.

Two conventions hold throughout this paper. A *mean* is always the arithmetic mean of individual timed runs, with warmup runs excluded. A *speedup* is always the ratio of two such means — never the mean of per-run or per-ordering ratios. The two are not equal, and every ratio we report is the former.

We prefer this to benchmarking separately-built identical binaries, which we also tried. Those would be measured in their own invocation, with a different command set and a different thermal profile from the runs they are meant to bound. The baseline’s spread is measured in the same invocations, on the same graphs, against the same competitors as the results it guards.

3.5 Phase instrumentation

Phase costs are measured inside the process. An identical patch — applied by cherry-pick, so all nine diffs match byte for byte — reads a monotonic clock around each phase and writes the elapsed time to standard error. It is off unless an environment variable is set, so layout output is unchanged and the correctness check below is unaffected.

Parsing and output writing happen outside the layout routine, so the phases do not sum to wall clock. We report the difference rather than assigning it to a phase.

3.6 Correctness checking

Correctness is checked on `dot -Tdot` output, not a rendered image. The text format is deterministic, so byte-identity means something; image comparison would fold in rendering and floating-point behaviour we are not studying. We also read both diffs against the prompt’s list of disallowed shortcuts.

3.7 Platform

Experiments ran on an Apple M4 Pro (12 cores: 8 performance, 4 efficiency), 24 GB unified memory, macOS 26.5.2, with Apple Clang 17.

4 Results

4.1 Noise floor

Because command order rotates, the baseline is measured once in each position. Table 2 reports how far apart its four means land — the apparent speedup this protocol produces when nothing has changed.

Worked example, crossing minimization on sparse-deep at 5,000 nodes: the four per-ordering means are 60.96, 61.36, 61.67 and 60.18 ms, giving six pairwise ratios between 1.005 and 1.025. The largest is the cell.

The floor is not uniform, which is why we report it per graph, per phase and per size. At 5,000 nodes the noisiest cell is coordinate assignment on dense-shallow at $1.155\times$, and it is not a property of that phase: the fourth ordering measures 295.0 ms against 255.4, 259.5 and 260.1 in the other three, and the same ordering also inflates that graph’s end-to-end figure. One slow ordering raises every cell it touches.

Which graph is noisiest is not stable either. Dense-shallow has the worst end-to-end floor at 5,000 nodes ($1.069\times$) and the best at 10,000 ($1.015\times$), while default goes the other way, from 1.033 to $1.089\times$. Nothing about a shape makes it intrinsically noisy; a floor describes the invocation it was

Table 2: Apparent speedup measured between the baseline and itself, per graph and per phase. **Each cell:** average the baseline’s 10 timed runs within each of the four command orderings, giving four means; form all six pairwise ratios, larger divided by smaller; the cell is the largest of the six.

Measurement	5,000 nodes			10,000 nodes		
	sparse	default	dense	sparse	default	dense
End-to-end	1.034	1.033	1.069	1.033	1.089	1.015
Initialization	1.022	1.029	1.090	1.020	1.059	1.012
Ranking	1.041	1.037	1.061	1.018	1.103	1.009
Crossing minimization	1.025	1.052	1.062	1.007	1.060	1.018
Coordinate assignment	1.039	1.078	1.155	1.047	1.142	1.027
Spline routing	1.055	1.030	1.066	1.005	1.039	1.040

measured in. That is the argument for measuring one inside every experiment rather than carrying a number across experiments, or across sizes.

Six ratios per cell is still a small sample, so each is a rough bound. That is the cost of not conflating graphs.

Throughout the rest of this section, we treat a measured ratio as a change only if it exceeds the floor for that same phase and that same graph. Ratios that do not are marked in the tables that follow.

This floor is measured inside the same *hyperfine* invocations as the results it bounds, on the same graphs, against the same competing commands. The more conventional construction — benchmarking separately-built identical binaries — places them in an invocation of their own, with a different command set and thermal profile, and so describes a different experiment from the one being guarded.

4.2 Phase weights depend on shape, not size

Which phase dominates is not fixed, and two things move it. At identical node count, shape moves it the most: crossing minimization takes 12.4% of layout time on one shape and 64.8% on another, while coordinate assignment moves the other way, 44.3% down to 9.2%. Size moves it too, and in a consistent direction — doubling to 10,000 nodes cuts crossing minimization’s share on every shape and hands the difference to network simplex. Each combination sets its own ceiling, which is why one optimization gives different end-to-end results on different graphs and at different sizes.

Table 3 and Figure 1 give the baseline cost of each phase.

Table 3: Baseline phase cost at 5,000 nodes. Phases are timed inside the process, so these are direct measurements rather than differences. **ms:** mean of that phase over all 40 timed runs (10 runs $\times$ 4 command orderings). **share:** that phase’s ms divided by the sum of the five phase ms. **Unaccounted:** wall clock minus that sum — parsing and output writing, which happen outside the layout routine.

Phase	sparse-deep		default		dense-shallow	
	ms	share	ms	share	ms	share
Initialization	114.8	23.3%	116.1	9.2%	118.8	4.1%
Ranking	75.7	15.3%	264.7	21.0%	532.5	18.4%
Crossing minimization	61.0	12.4%	571.7	45.4%	1878.5	64.8%
Coordinate assignment	218.7	44.3%	240.9	19.1%	267.5	9.2%
Spline routing	23.3	4.7%	66.3	5.3%	99.5	3.4%
Measured total	493.6	100%	1259.7	100%	2896.7	100%
Wall clock	557.5		1360.7		3029.1	
Unaccounted	64.0	11.5%	101.0	7.4%	132.4	4.4%

We report the unaccounted row rather than folding it into a phase. It is largest on sparse-deep, where the layout work is smallest and the fixed cost of parsing shows through, and it shrinks at 10,000 nodes on every shape because the layout work grows while parsing and output do not.

Table 4: Baseline phase cost at 10,000 nodes, measured the same way. The shapes are generated with the same nodes-per-rank and fanout as their 5,000 counterparts, so only the number of ranks doubles.

Phase	sparse-deep		default		dense-shallow	
	ms	share	ms	share	ms	share
Initialization	134.9	7.2%	138.1	3.1%	136.9	1.3%
Ranking	279.1	15.0%	1262.9	28.4%	3083.9	29.5%
Crossing minimization	152.7	8.2%	1382.9	31.1%	5458.7	52.3%
Coordinate assignment	1245.7	66.9%	1511.4	34.0%	1547.5	14.8%
Spline routing	48.6	2.6%	148.6	3.3%	214.9	2.1%
Measured total	1861.0	100%	4444.0	100%	10441.9	100%
Wall clock	1975.9		4643.4		10699.3	
Unaccounted	114.9	5.8%	199.5	4.3%	257.4	2.4%

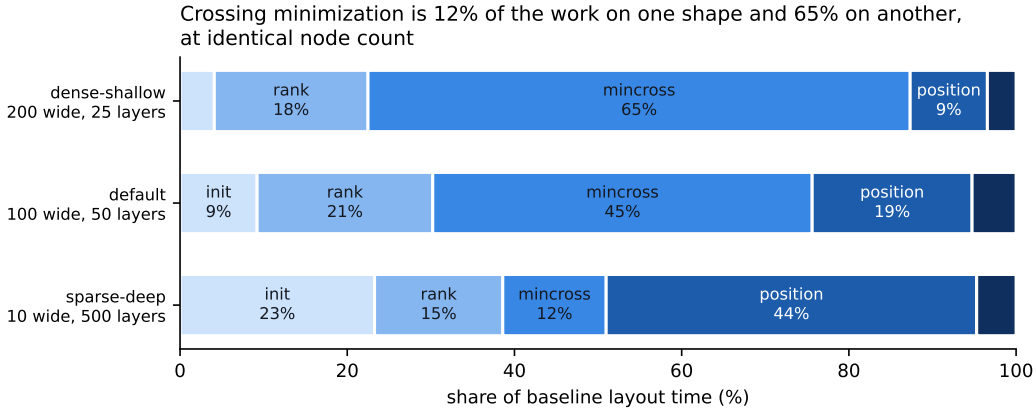


Figure 1: Baseline runtime composition by shape at 5,000 nodes.

Doubling the node count moves the weights in one direction on all three shapes. Crossing minimization’s share falls — 12.4% to 8.2%, 45.4% to 31.1%, 64.8% to 52.3% — and network simplex takes what it gives up. On sparse-deep, coordinate assignment goes from 44.3% to 66.9% of layout time; on dense-shallow, ranking nearly doubles its share. Both network simplex phases grow faster than linearly in node count while crossing minimization, whose cost is dominated by within-rank comparisons at fixed rank width, grows closer to linearly.

So composition is not a property of shape alone. At a fixed size, shape separates the three workloads sharply; across sizes, every shape drifts the same way. Any claim about which phase is worth optimizing has to name a size as well as a shape.

4.3 End-to-end results

Table 5 and Figure 2 give layout time for all nine binaries on each shape.

At 5,000 nodes all four Claude runs improve all three shapes, from $1.39\times$ to $2.91\times$, and every one of those twelve ratios clears its floor by a wide margin. Among the four Codex runs, only the first shows any improvement at all: $1.16\times$ on default and $1.33\times$ on dense-shallow, with sparse-deep inside the floor. The other three Codex runs are indistinguishable from the unmodified baseline on every graph.

That is not a failure to optimize. Three of those four Codex runs shipped no layout code, by their own decision, and Section 4.6 treats them separately — both because averaging them with the first would describe a binary nobody produced, and because a binary that is the baseline recompiled tells us something useful about the floor.

Table 5: End-to-end layout time in milliseconds at 5,000 nodes. **Mean and sd:** over all 40 timed runs (10 runs $\times$ 4 command orderings). **Speedup:** baseline mean divided by that binary’s mean. A dagger marks a ratio that does not exceed the end-to-end floor for that graph in Table 2; we read those as no change. **P** marks a run given the parallelism prompt.

Binary	sparse-deep		default		dense-shallow	
Baseline	557.5 $\pm$ 9.6		1360.7 $\pm$ 40.0		3029.1 $\pm$ 99.9	
Claude 1	399.9 $\pm$ 3.2	(1.39)	878.4 $\pm$ 27.7	(1.55)	1603.3 $\pm$ 64.3	(1.89)
Claude 2	334.6 $\pm$ 3.6	(1.67)	685.7 $\pm$ 20.1	(1.98)	1288.7 $\pm$ 31.9	(2.35)
Claude 1 P	349.3 $\pm$ 4.1	(1.60)	646.8 $\pm$ 26.2	(2.10)	1102.6 $\pm$ 16.1	(2.75)
Claude 2 P	378.2 $\pm$ 7.0	(1.47)	620.8 $\pm$ 30.3	(2.19)	1042.1 $\pm$ 20.7	(2.91)
Codex 1	543.1 $\pm$ 8.5	(1.03) [†]	1172.7 $\pm$ 36.5	(1.16)	2284.8 $\pm$ 62.5	(1.33)
Codex 2	559.5 $\pm$ 7.6	(1.00) [†]	1325.4 $\pm$ 49.2	(1.03) [†]	3026.6 $\pm$ 154.2	(1.00) [†]
Codex 1 P	555.8 $\pm$ 11.0	(1.00) [†]	1336.9 $\pm$ 72.0	(1.02) [†]	2974.4 $\pm$ 80.3	(1.02) [†]
Codex 2 P	554.6 $\pm$ 10.4	(1.01) [†]	1351.8 $\pm$ 108.7	(1.01) [†]	3002.7 $\pm$ 115.8	(1.01) [†]

Table 6: End-to-end layout time in milliseconds at 10,000 nodes, same protocol and same dagger convention, read against the 10,000-node floors in Table 2.

Binary	sparse-deep		default		dense-shallow	
Baseline	1975.9 $\pm$ 46.1		4643.4 $\pm$ 266.2		10699.3 $\pm$ 80.8	
Claude 1	1216.7 $\pm$ 20.5	(1.62)	2941.0 $\pm$ 176.9	(1.58)	5510.8 $\pm$ 62.6	(1.94)
Claude 2	782.0 $\pm$ 7.4	(2.53)	1837.6 $\pm$ 41.3	(2.53)	4018.2 $\pm$ 43.8	(2.66)
Claude 1 P	831.3 $\pm$ 7.0	(2.38)	1493.3 $\pm$ 58.3	(3.11)	3094.1 $\pm$ 18.2	(3.46)
Claude 2 P	1045.6 $\pm$ 8.7	(1.89)	1414.3 $\pm$ 26.0	(3.28)	2660.2 $\pm$ 16.2	(4.02)
Codex 1	1928.7 $\pm$ 36.1	(1.02) [†]	4080.2 $\pm$ 181.6	(1.14)	8654.4 $\pm$ 54.5	(1.24)
Codex 2	1967.5 $\pm$ 25.1	(1.00) [†]	4436.0 $\pm$ 175.7	(1.05) [†]	10710.6 $\pm$ 78.2	(1.00) [†]
Codex 1 P	1953.6 $\pm$ 21.4	(1.01) [†]	4490.0 $\pm$ 214.4	(1.03) [†]	10701.5 $\pm$ 116.2	(1.00) [†]
Codex 2 P	1955.3 $\pm$ 24.1	(1.01) [†]	4697.2 $\pm$ 417.5	(0.99) [†]	10727.5 $\pm$ 102.3	(1.00) [†]

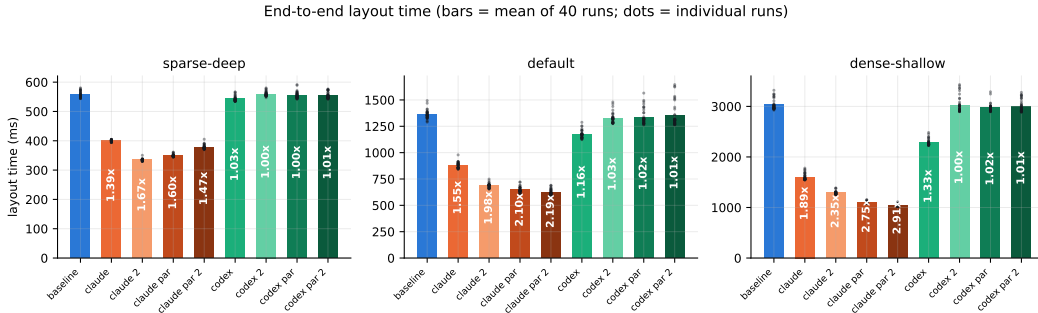


Figure 2: End-to-end layout time on all three shapes at 5,000 nodes.

The spread between repetitions of the same prompt is larger than we expected. Claude’s two base-prompt runs differ by 1.39 against 1.67, 1.55 against 1.98, and 1.89 against 2.35 — the second run is faster on every shape, by more than the gap between some agents. Codex’s two base-prompt runs differ more sharply still: one shipped two changes, the other shipped nothing.

At 10,000 nodes every run that optimized network simplex improves by more (Table 6), which follows from the composition shift: the phases these runs optimized are a larger share of a larger graph. The ordering does not simply scale, though. Claude’s second parallelism run is the fastest binary we measured on dense-shallow at $4.02\times$, and the *slowest* of the four Claude runs on sparse-deep at $1.89\times$, where crossing minimization — the phase its concurrency targets — is 8.2% of the work. The gap between the two base-prompt Claude runs widens with size: they are 1.39 against 1.67 at 5,000 on sparse-deep and 1.62 against 2.53 at 10,000.

Codex’s one shipping run moves the other way. Its default-shape result falls from 1.16 to $1.14\times$ while the floor for that cell rises to $1.089\times$, so it now barely clears, and its dense-shallow result falls from 1.33 to $1.24\times$. An optimization confined to crossing minimization is worth less on bigger graphs, because that is the phase whose share shrinks with size — the ceiling it works under drops from 2.63 to $2.04\times$ on that shape.

4.4 What each run changed

Table 7 summarizes each run’s diff and report: where it changed code, what the main change was, whether and how it used threads, and whether the change survived to the binary we measured.

Table 7: What each run changed, from its commit history and report. **Concurrency:** thread model; default thread count; input-size gates below which the code stays serial; how output is kept independent of scheduling. **E2E:** smallest and largest end-to-end speedup across all six graphs (Tables 5 and 6); for Codex 1 the lower end is inside the floor. **P** marks the parallelism prompt. **NS:** network simplex.

Run	Code changed	Main change	Concurrency	Outcome	E2E
Claude 1	<code>ns.c</code> (<code>dfs_range</code> , <code>enter_edge</code> , <code>rerank</code> , <code>init_graph</code>); <code>mincross.c</code> (<code>transpose_step</code>)	NS over a flat per-node state array with cached tree neighbours; both crossing orderings counted in one pass	none	shipped	1.39– 1.94 $\times$
Claude 2	<code>ns.c</code> (whole solver); <code>mincross.c</code> (<code>transpose</code>)	NS on compact index arrays with preallocated DFS stacks; crossings counted on flattened edge lists; one experiment (gap-spaced labels) reverted	none	shipped	1.67– 2.66 $\times$
Claude 1 P	<code>ns.c</code> ; <code>mincross.c</code> ; new <code>lib/util/parallel.c</code>	NS subtree relabelling via a postorder index array; a rank’s edge ends snapshot once per transpose step	pthread fork-join pool; all cores (12); NS loops on subtrees $\geq 16,384/32,768$ nodes; <code>mincross</code> passes 0 and 1 run concurrently on private copies; disjoint writes and a deterministic merge	shipped	1.60– 3.46 $\times$
Claude 2 P	<code>ns.c</code> ; <code>mincross.c</code> ; new <code>lib/common/parallel.c</code>	NS subtree relabelling via an inverse label array	persistent pool on C11 atomics; performance cores (8); <code>transpose</code> as a wavefront on graphs $\geq 2,000$ nodes, with an undo log for speculative sweeps; NS scans on subtrees $\geq 16,384/32,768$ nodes	shipped	1.47– 4.02 $\times$
Codex 1	<code>mincross.c</code> (<code>transpose_step</code> , <code>rcross</code>)	Two crossing-pair scans fused into one; Fenwick-tree prefix sums in <code>rcross</code>	none	shipped	1.02– 1.33 $\times$
Codex 2	<code>mincross.c</code> (<code>rcross</code>)	Fenwick-tree crossing counter	none	reverted: 17% slower at 25,000 nodes	$\approx 1.00\times$
Codex 1 P	<code>mincross.c</code> (<code>ncross</code>)	Adjacent-rank crossing counts computed concurrently	OpenMP; private result slots summed in rank order	reverted: $0.92\times$ at 2 threads	$\approx 1.00\times$
Codex 2 P	<code>mincross.c</code> (<code>ncross</code>)	Same as Codex 1 P	threads; same merge	reverted: gain confined to a small phase	$\approx 1.00\times$

4.5 Per-phase attribution

Table 8 and Figure 3 break the end-to-end result down by phase.

Table 8: Per-phase speedup over baseline at 5,000 nodes, for the five runs that shipped layout code. **Each entry:** the baseline’s mean for that phase divided by that binary’s mean for the same phase, each mean taken over all 40 timed runs. A dagger marks an entry that does not exceed the floor for that same phase and graph in Table 2; those are no change, not small effects. **P** marks a run given the parallelism prompt.

Phase	Claude 1	Claude 2	Claude 1 P	Claude 2 P	Codex 1
<i>sparse-deep</i>					
Initialization	1.01 [†]	1.01 [†]	1.00 [†]	1.00 [†]	1.00 [†]
Ranking	2.55	2.85	4.59	4.47	0.99 [†]
Crossing minimization	1.48	1.84	2.50	1.26	1.32
Coordinate assignment	1.70	2.92	2.06	2.00	1.00 [†]
Spline routing	1.01 [†]	1.03 [†]	1.01 [†]	0.99 [†]	1.01 [†]
<i>default</i>					
Initialization	1.00 [†]	1.00 [†]	0.97 [†]	0.97 [†]	0.98 [†]
Ranking	2.07	2.53	4.36	4.19	0.99 [†]
Crossing minimization	1.91	2.67	3.28	3.85	1.53
Coordinate assignment	1.42	2.84	1.97	2.05	0.99 [†]
Spline routing	1.01 [†]	1.01 [†]	1.00 [†]	0.98 [†]	1.00 [†]
<i>dense-shallow</i>					
Initialization	1.01 [†]	1.01 [†]	1.02 [†]	1.01 [†]	1.02 [†]
Ranking	2.10	2.50	4.57	4.30	1.01 [†]
Crossing minimization	2.37	2.95	3.71	4.31	1.64
Coordinate assignment	1.28	2.89	1.98	2.03	1.02 [†]
Spline routing	1.01 [†]	1.02 [†]	1.03 [†]	1.00 [†]	1.01 [†]

Table 9: Per-phase speedup over baseline at 10,000 nodes, same convention, read against the 10,000-node floors in Table 2.

Phase	Claude 1	Claude 2	Claude 1 P	Claude 2 P	Codex 1
<i>sparse-deep</i>					
Initialization	1.00 [†]	1.01 [†]	1.00 [†]	1.01 [†]	0.99 [†]
Ranking	2.79	3.10	6.03	5.90	0.98 [†]
Crossing minimization	1.47	1.93	2.61	1.58	1.34
Coordinate assignment	1.74	3.95	2.91	2.06	1.01 [†]
Spline routing	1.00 [†]	1.01 [†]	1.00 [†]	1.00 [†]	1.00 [†]
<i>default</i>					
Initialization	0.99 [†]	1.00 [†]	0.98 [†]	1.02 [†]	1.01 [†]
Ranking	2.15	2.64	5.44	5.40	1.02 [†]
Crossing minimization	1.82	2.53	3.31	4.35	1.50
Coordinate assignment	1.37	4.60	4.29	3.97	1.05 [†]
Spline routing	0.99 [†]	1.01 [†]	1.00 [†]	1.01 [†]	1.01 [†]
<i>dense-shallow</i>					
Initialization	1.01 [†]	1.01 [†]	1.00 [†]	0.98 [†]	1.00 [†]
Ranking	2.17	2.56	5.52	5.21	0.99 [†]
Crossing minimization	2.34	2.89	3.53	5.10	1.61
Coordinate assignment	1.34	4.85	4.08	4.11	1.00 [†]
Spline routing	1.00 [†]	1.01 [†]	1.00 [†]	0.97 [†]	1.00 [†]

Thirty-nine of the 75 entries exceed their floor, and they fall in an obvious pattern: every Claude run on ranking, crossing minimization and coordinate assignment on all three graphs, plus Codex’s first run on crossing minimization on all three. Initialization and spline routing never clear anywhere, which is the control we expect — no run touched either.

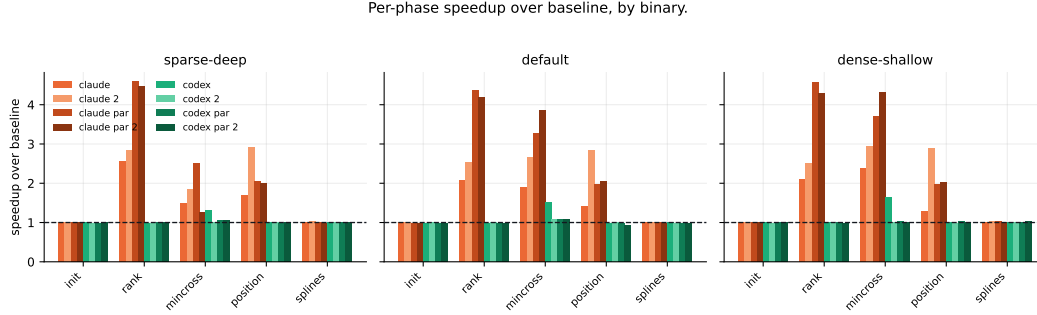


Figure 3: Per-phase speedup at 5,000 nodes. Every Claude run improves ranking, crossing minimization and coordinate assignment on every shape; the one Codex run that shipped code improves crossing minimization only.

The four Claude runs divide the same three phases differently. Ranking is where the parallelism-prompt runs separate themselves: 4.19 to $4.59\times$ against 2.07 to $2.85\times$ for the base-prompt runs. Crossing minimization ranges from 1.26 to $4.31\times$, and coordinate assignment from 1.28 to $2.92\times$, with no run dominating the others on all three.

One entry is worth isolating. Claude’s second parallelism run manages $4.31\times$ on crossing minimization on dense-shallow but only $1.26\times$ on sparse-deep — the widest spread any run shows on any phase. Its concurrency is in transpose, which needs rank width to have anything to divide, and sparse-deep is ten nodes wide.

Codex’s first run improves crossing minimization by 1.32 to $1.64\times$ and nothing else. That is what produces its $1.03\times$ end-to-end on sparse-deep, inside the floor: the optimization is real on the phase, but the phase is 12.4% of that workload.

At 10,000 nodes (Table 9) the same pattern holds with every effect larger. The two parallelism runs reach 5.21 to $6.03\times$ on ranking, against 2.15 to $3.10\times$ for the base-prompt runs. Coordinate assignment separates the runs most: Claude’s second base-prompt run reaches 3.95 to $4.85\times$ there — the best of any run on that phase — while Claude’s first manages 1.34 to $1.74\times$. Codex’s shipping run still improves crossing minimization and nothing else, now by 1.34 to $1.61\times$.

Two entries move enough to be worth naming. Claude’s second parallelism run goes from 1.26 to $1.58\times$ on crossing minimization on sparse-deep and from 4.31 to $5.10\times$ on dense-shallow, widening the gap its rank-width dependence creates. And every run’s coordinate-assignment speedup grows with size, including the runs that ship no threads at all, which matters for attributing the parallelism runs’ gains below.

4.6 The runs that shipped nothing

Three of the four Codex runs changed no layout code at all. Their complete diff against the starting commit is the report each was asked to write, plus, in one case, an unused benchmark generator under `tools/`. Under `lib/` and `cmd/` the diff is empty. Those three binaries are the baseline, recompiled.

Each arrived there by a different route, and in each case by reverting work it had already committed:

- **Codex 2** replaced the crossing counter with a Fenwick tree, then measured a 17% regression at 25,000 nodes — the shape the prompt named — and reverted it. Its report attributes the loss to the original scan being cache-friendly at these rank widths.
- **Codex 1 P** parallelized adjacent-rank crossing counts with OpenMP, measured $0.92\times$ at two threads, and reverted.
- **Codex 2 P** did the same with pthreads, and reverted on the grounds that the phase it could parallelize was 0.2 s against a positioning phase of 1.7 to 3.0 s.

We report these as deliberate outcomes, not as failed optimizations. Each run built something, measured it against the workload the prompt described, and concluded the change did not earn its place. The prompt explicitly allowed that outcome.

They also give us something we did not design for: three binaries that are byte-for-byte the baseline in every line of layout code, measured in the same invocations as the rest, under different names and in different command positions. Their nine end-to-end ratios run from 1.00 to $1.03\times$, which is what the floor predicts.

Their per-phase entries are less reassuring. All three measure 1.07 to $1.08\times$ on crossing minimization on sparse-deep and default — above the 1.025 and 1.052 floors for those cells — and one measures $0.93\times$ on coordinate assignment on default. Three independent binaries containing no change do not agree on a $1.07\times$ improvement by chance, so this is a position effect the baseline-against-itself construction does not capture: the floor in Table 2 compares the baseline only with copies of itself run in the same invocation, and cross-binary variation is evidently larger.

The practical consequence for this paper is small, because every result we claim clears its floor several times over. But a per-phase ratio near $1.1\times$ in this protocol should not be read as an effect, and we would not report one as such.

4.7 How much of the parallel gain is threads

The two parallelism-prompt runs from Claude both ship thread pools, and both choose their own thread count. Running them again at `GV_THREADS=1` separates what their concurrency buys from what their serial changes buy. Figure 4 reports that measurement at both sizes, taken in its own hyperfine invocations with two orderings and 20 timed runs per configuration, so its ratios should be read against its own baselines (578.2, 1368.3, 3038.9 ms at 5,000 nodes; 1925.4, 4406.7, 10925.6 ms at 10,000) rather than against Tables 5 and 6.

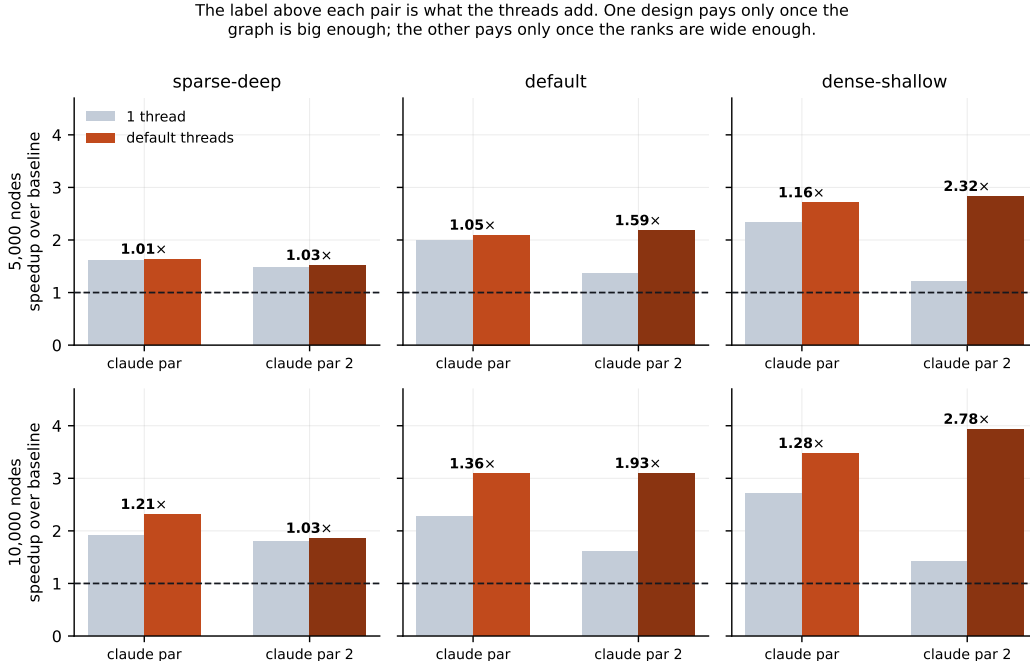


Figure 4: Each binary at one thread and at its own default, as speedup over baseline, at both node counts. The label above each pair is the default-thread mean divided by the one-thread mean — the part of the speedup that concurrency accounts for. Claude 1 P’s threads are inert at 5,000 nodes and contribute at 10,000; Claude 2 P’s contribute wherever ranks are wide, at either size.

The two runs are not the same kind of result, despite landing close together in Table 5, and which one is a parallel result depends on the input.

At 5,000 nodes, Claude’s first parallelism run is a serial optimization that also contains threads. At one thread it already reaches 1.61, 1.99 and 2.33 $\times$; enabling its threads adds 1 to 16%. Its report explains why: its concurrency sits inside network simplex behind subtree-size thresholds of 32,768 and 16,384 nodes, and those paths rarely execute on a 5,000-node graph. What it gained, it gained from rewriting subtree relabelling as array operations.

At 10,000 nodes the same binary’s threads begin to pay: 1.21, 1.36 and 1.28 $\times$, against 1.01, 1.05 and 1.16 $\times$ at half the size. The coordinate-assignment phase solves an auxiliary graph several times larger than the input, so at this size it crosses the thresholds the agent chose. It is still predominantly a serial result — at one thread it reaches 1.91 to 2.71 $\times$ — but the concurrency it shipped is no longer inert.

Claude’s second parallelism run is the opposite at both sizes. At one thread it is the weaker of the two everywhere. With threads it becomes the fastest binary we measured, and on dense-shallow concurrency accounts for 2.32 $\times$ of its result at 5,000 nodes and 2.78 $\times$ at 10,000. Its threading is in transpose, with a 2,000-node threshold every graph here clears, and transpose is most of crossing minimization.

Sparse-deep separates the two designs most sharply, and it is the clearest result in this section. At 10,000 nodes the first run gets 1.21 $\times$ from threads there and the second gets 1.03 $\times$ — the reverse of dense-shallow, where they get 1.28 and 2.78 $\times$. The first put its concurrency in network simplex, which is 66.9% of sparse-deep’s runtime at that size; the second put it in crossing minimization, which is 8.2%. Neither design is better. Each pays exactly when the input makes its phase expensive, which is the same composition argument that governs the serial results.

This is the measurement that decides how the parallelism prompt should be described. It produced two runs that look equally parallel end to end, and whether either deserves the label depends on the graph you hand it.

4.8 Correctness

All eight binaries produce `dot -Tdot` output byte-identical to the baseline on every evaluation graph, at both node counts and with instrumentation both enabled and disabled: 96 comparisons, no differences. The baseline reproduces itself on all six graphs, so the comparison is meaningful on each.

We also ran the 260 graphs in GraphViz’s own test suite through the five binaries that shipped layout code. Every run changed `mincross.c`, which only `dot` uses; the four Claude runs also changed `ns.c`, which `neato` reaches as well, so we checked both engines. All 2,600 comparisons are identical, with matching exit codes and nothing skipped. We do not report `fdp` or `circo`: the baseline `fdp` is not reproducible against itself on these inputs, so byte comparison is meaningless for it, and baseline `circo` does not finish one test graph within an hour.

Two of the binaries create threads, so byte-identity at one thread count is not sufficient for them. We repeated the whole test-suite comparison for both at `GV_THREADS` of 1, 2 and 8 and at their own default: 2,080 further comparisons, all identical.

That check on its own would prove less than it appears to. Both binaries gate their concurrency behind input-size thresholds — 16,384- and 32,768-node subtrees in one, 2,000 nodes in the other — and almost every graph in `tests/graphs` is far below all of them, so those 2,080 comparisons largely exercise the serial paths. We therefore repeated the comparison on the three evaluation graphs, which are large enough to execute the parallel code: each binary at `GV_THREADS` of 1, 2, 4, 8, 12 and its own default, three repetitions each, for 108 comparisons. All are identical to the baseline.

Output does not depend on thread count in anything we ran, which is evidence for the agents’ determinism arguments but not a proof of race-freedom — see Section 6.

The remaining three binaries need no such check. Their layout sources are identical to the baseline’s (Section 4.6), so their output is identical by construction; they are nonetheless included in the 48 comparisons above.

Reading all eight diffs against the prompt’s disallowed shortcuts found no reduced iteration counts, early exits, input special-casing, cached layout results or modified tests. Two binaries do introduce concurrency, which the parallelism prompt asked for and which the same prompt required to leave output unchanged; the byte-identity results above are what we have to show for that requirement.

5 Discussion

5.1 Finding the bottleneck was not the hard part

No run was told where to look, and none of the eight failed to find expensive code. Every one profiled first, and every report names the same two regions: crossing minimization, and the network simplex that serves both ranking and coordinate assignment. Several runs independently arrived at the same transformation — fusing the two symmetric crossing-count passes into one. Locating the work is evidently not what separates these runs.

Breadth is. All four Claude runs went into network simplex as well as crossing minimization, and that is where their margin comes from: ranking and coordinate assignment improve in every Claude run and in no Codex run (Table 8). Codex’s one shipping run stopped at the phase it identified first. The pattern holds across three workloads of very different composition, so it is not an inference from a single measurement.

The reports also show the two agents disagreeing about the same code rather than seeing different things. Both Codex parallelism runs argue that transpose cannot be parallelized without changing output, because each accepted swap changes the state the next comparison reads, and that network simplex pivots are inherently sequential. Both Claude parallelism runs worked around those dependencies instead. One parallelized transpose itself, with a speculative sweep schedule and an undo log. The other left transpose serial, judging that most speculation would fail, and ran crossing minimization’s two independent initial passes concurrently on private copies. Both produced byte-identical output. The disagreement is not about where the time goes. It is about whether a dependency is a hard barrier or an engineering problem.

5.2 Agents optimize the workload they imagine

No run could see our evaluation graphs, so each generated its own. Their graphs ranged from 5,000 to 25,000 nodes, against our 5,000 and 10,000.

That choice has consequences we can measure. Claude’s first parallelism run put its concurrency behind subtree-size thresholds of 32,768 and 16,384 nodes, tuned on graphs where network simplex dominates. At 5,000 nodes those paths almost never execute, and its threads contribute between 1 and 16% (Figure 4) — there it scores well on the strength of a serial rewrite that happens to travel. Claude’s second parallelism run put its concurrency in transpose behind a 2,000-node threshold, which even our smallest graphs clear, and its threads contribute up to $2.32\times$.

Neither run was tuned to our benchmark; one simply guessed a workload closer to it. A study that reported only end-to-end numbers would have called both of them parallel wins.

At 10,000 nodes the first run improves sharply — 1.60 to $2.38\times$ on sparse-deep, 2.75 to $3.46\times$ on dense-shallow — and re-measuring it at one thread shows where that came from. Its threads, worth 1 to 16% at 5,000 nodes, are worth 21 to 36% at 10,000 (Figure 4). The concurrency it shipped was not useless; it was written for graphs larger than the ones we first measured it on, and it switches on when the input reaches the size it was tuned for. Most of its gain is still serial even at 10,000 nodes, so the description of it as a serial optimization that also contains threads survives — but only as a statement about a size range, not about the binary.

The reverse case is sharper. On sparse-deep at 10,000 nodes the first run’s threads are worth $1.21\times$ and the second run’s $1.03\times$; on dense-shallow the same comparison is 1.28 against $2.78\times$. One agent put its concurrency in network simplex and the other in crossing minimization, and each is rewarded exactly on the shapes where its phase is expensive. Two agents given the same instruction to use the idle cores produced designs whose value is decided by the input, not by the instruction.

5.3 Why breadth mattered: the ceiling moves with shape

Amdahl’s law [1] bounds what optimizing one phase can yield, and that bound depends on the phase’s share of runtime — which Table 3 shows is a property of the graph’s shape.

Table 10: What optimizing crossing minimization alone can achieve, against what each run measured. **Ceiling:** baseline wall clock divided by (wall clock minus that phase’s measured cost), i.e. the phase driven to zero. Remaining columns are the end-to-end speedups from Tables 5 and 6, carrying the same dagger convention. Entries above the ceiling are bold.

Graph	Ceiling	Claude 1	Claude 2	Claude 1 P	Claude 2 P	Codex 1
<i>5,000 nodes</i>						
sparse-deep	1.12×	1.39	1.67	1.60	1.47	1.03 [†]
default	1.72×	1.55	1.98	2.10	2.19	1.16
dense-shallow	2.63×	1.89	2.35	2.75	2.91	1.33
<i>10,000 nodes</i>						
sparse-deep	1.08×	1.62	2.53	2.38	1.89	1.02 [†]
default	1.42×	1.58	2.53	3.11	3.28	1.14
dense-shallow	2.04×	1.94	2.66	3.46	4.02	1.24

Codex’s one shipping run sits below the ceiling everywhere, as a run that optimized only that phase must. On sparse-deep at 5,000 nodes the ceiling is 1.12×, so even a perfect crossing-minimization optimization would be nearly invisible end-to-end. Its 1.03× is not a failed optimization — the change works, at 1.32× on the phase — but the phase is worth 12.4% of that workload.

Every Claude run *exceeds* that ceiling on sparse-deep, at both sizes, which is only possible by improving other phases. That is the breadth argument at its sharpest: on that workload no amount of further work on the phase every run found could have matched what they did.

The ceiling also explains why the shapes rank the runs differently. On dense-shallow at 5,000 nodes, where crossing minimization is 64.8% of the work, the ceiling is 2.63× and only the two parallelism runs clear it — breadth matters less there, because the one phase everybody found is most of the workload. On sparse-deep the ceiling is so low that breadth is the only thing that matters.

Doubling the node count lowers every ceiling: 1.12 to 1.08×, 1.72 to 1.42×, 2.63 to 2.04×. Crossing minimization is a smaller share of a larger graph, so the strategy of optimizing it alone is worth less exactly where the absolute times are worst. Three Claude runs clear the dense-shallow ceiling at 10,000 nodes where only two did at 5,000, and on default every Claude run now clears a ceiling that has dropped below what three of them already achieved. Breadth is not merely a way to go faster here; it is the only strategy whose value grows with the problem.

A single benchmark graph would have hidden all of this — and a single graph *size* would have hidden the direction it moves in.

The ceiling an optimization can reach is a property of the input, not the program, and a single benchmark graph cannot show that.

5.4 Cost

Table 11: Cost of each Claude run, as reported by its session log. All four ran `claude-opus-5`. **P** marks the parallelism prompt.

Metric	Claude 1	Claude 2	Claude 1 P	Claude 2 P
Working time	4 h 27 m	1 h 16 m	3 h 23 m	5 h 22 m
Input tokens	730	2,838	3,914	3,848
Output tokens	620,448	274,861	229,374	212,306
Cache creation	1,270,669	437,852	805,875	667,657
Cache read	89,851,473	26,712,125	36,947,373	40,939,987
Total tokens	91,743,320	27,427,676	37,986,536	41,823,798

Table 12: Cost of each Codex run, as reported by its session log. All four ran gpt-5.6-terra. Cached input is a subset of input; reasoning output is a subset of output. **P** marks the parallelism prompt.

Metric	Codex 1	Codex 2	Codex 1 P	Codex 2 P
Working time	23 m 42 s	12 m 55 s	9 m 50 s	9 m 30 s
Input tokens	5,114,035	2,406,206	1,684,133	1,634,219
of which cached	4,985,600	2,313,728	1,592,064	1,559,808
Output tokens	20,826	17,782	15,144	13,443
of which reasoning	4,555	4,101	5,684	3,734
Cache-write tokens	0	0	0	0
Total tokens	5,134,861	2,423,988	1,699,277	1,647,662

The gap is consistent and large. Claude’s runs took between 1 and 5 hours and between 27 and 92 million tokens; Codex’s took between 9 and 24 minutes and between 1.6 and 5.1 million. The cheapest Claude run cost roughly five times the most expensive Codex run in tokens, and about three times in wall time.

Both columns move together within each agent, and neither predicts the result. Claude’s most expensive run by tokens is its first, which is also its slowest binary of the four; its longest run by time is the one that produced the fastest binary. Codex’s two cheapest runs shipped nothing, but so did its second run, which cost twice as much.

These are not like-for-like counts. The two systems report different categories — one folds cached reads into a total that dwarfs everything else, the other reports cache reads as a subset of input — and a token is not a unit of work. We report them because they are what the logs record, and because the order-of-magnitude difference is too large to leave out, not because they support a cost-per-speedup comparison.

6 Limitations

We state these plainly because several of them bound the conclusions above.

Two runs per prompt is still a small sample. Each agent ran twice under each prompt. That is enough to show the outcome moves — Claude’s two base-prompt runs differ by more than some agent-to-agent gaps, and Codex’s differ in whether any code shipped at all — but not enough to estimate how often either outcome occurs. Two runs can show that a single run is not characteristic. They cannot tell you what is, and nothing here should be read as a ranking of the two agents in general.

Our Codex result rests on one shipping run. Three of Codex’s four runs shipped no layout code, so every Codex speedup in this paper comes from a single run. We can say what that run achieved and what the other three decided; we cannot say what a typical Codex result looks like, because we have one sample of it.

Thread counts were barely explored. The main measurements give each binary whatever thread count it chooses for itself, which is what a user would get, but means the two threaded binaries are measured at different widths (all cores against performance cores only). Section 4.7 re-measures them at one thread and at their default, at both node counts, end to end only. Two points on a curve is enough to show that one design’s concurrency switches on between 5,000 and 10,000 nodes, but not to locate where, and we did not instrument which parallel paths actually executed. We also did not sweep intermediate thread counts, measure per-phase scaling, or test on a machine with a different core mix, so we cannot say where either design stops scaling or how it behaves when cores are contended.

We have not verified the absence of data races. Two of the binaries we measured create threads. Our evidence that they are safe is byte-identical output across 2,188 comparisons that vary thread count — the 260 test graphs at one, two, eight and default threads, and the three evaluation graphs

at one, two, four, eight, twelve and default threads with three repetitions each — plus the safety arguments in the agents’ own reports.

That is consistent with race-freedom and is not proof of it, for three reasons. Byte-identical output is an end-state check: a race that corrupts state the layout later overwrites, or that loses to the same winner every time on this hardware, produces identical bytes. Three repetitions per configuration is a thin sample for a timing-dependent fault. And every run was on one machine with one core mix, so schedules that only occur under contention, on more cores, or on a weaker memory model were never attempted. Both Claude parallelism runs report that ThreadSanitizer could not be made to run on this machine, and we did not run one either. Running these inputs under a dynamic race detector on Linux is the check this paper is missing.

Synthetic graphs, and only two sizes. All three shapes come from the same generator, which produces uniform rank widths and constant fanout. Real inputs to `dot` are irregular in both. Two node counts are enough to show that composition drifts with size and which direction it drifts, but not to fit a curve: we cannot say whether crossing minimization’s share keeps falling at 50,000 nodes or levels off. Several agents generated their own test graphs at up to 25,000 nodes, larger than anything we measure, and at least one run’s concurrency is gated behind subtree-size thresholds that our 5,000-node graphs never reach and our 10,000-node graphs may (Section 4.7). Our numbers describe these binaries on these workloads; a third size could reorder them again.

The noise floor understates cross-binary variation. Section 4.6 shows three binaries containing no layout change measuring up to $1.08\times$ on a phase whose floor we computed as $1.03\text{--}1.05\times$. The floor is built from the baseline against copies of itself, and evidently does not capture everything that varies between separately named commands in one invocation. Every result we claim clears its floor several times over, but a per-phase ratio near $1.1\times$ in this protocol should not be read as an effect.

One machine, one toolchain, one point in time. Every measurement is from a single Apple M4 Pro running macOS 26.5.2 and Apple Clang 17. The wins the agents found are largely memory-hierarchy wins, and their size will differ on a machine with a different cache hierarchy or core mix. The agents themselves are moving targets: these are specific model versions at specific dates, and nothing here predicts what the same prompt produces later.

Reproducibility Statement

Every number in this paper comes from data in the artifact below. No result is estimated, extrapolated, or reported from a source other than the raw measurement files.

Artifacts. Everything except the agents’ own work lives in one repository, <https://github.com/harvest7777/paper-rewrite>:

- `prompts/` — both prompts, verbatim as the agents received them.
- `graphs/` — the fixed-seed generator and the three evaluation graphs it produces.
- `new_paper/` — this paper’s source, and `figures/` within it, which regenerates every figure from the raw measurements.
- `01-build-binaries.sh`, `02-run.sh`, `03-test.sh`, `04-thread-scaling.sh` — the build, benchmark, correctness and thread-decomposition harnesses, numbered in the order they are run.
- `results/` — every raw measurement behind this paper: the per-run hyperfine JSON, the per-run phase logs in `phases/`, the thread-scaling run in `thread-scaling/`, the correctness sweeps and their raw output in `correctness/`, and one session dump per run in `dumps/`.

Each agent run has its own repository, because each was a separate GraphViz checkout. The link gives that run’s complete diff against the starting commit, together with the commit history the agent produced and the report it wrote.

- Claude, run 1:
<https://github.com/harvest7777/claude-graphviz-run/compare/main...agent-work>
- Claude, run 2:
<https://github.com/harvest7777/claude-2-graphviz-run/compare/main...agent-work>
- Claude, parallelism prompt, run 1:
<https://github.com/harvest7777/claude-parallel-graphviz-run/compare/main...agent-work>
- Claude, parallelism prompt, run 2:
<https://github.com/harvest7777/claude-parallel-2-graphviz-run/compare/main...agent-work>
- Codex, run 1:
<https://github.com/harvest7777/codex-graphviz-run/compare/main...agent-work>
- Codex, run 2:
<https://github.com/harvest7777/codex-2-graphviz-run/compare/main...agent-work>
- Codex, parallelism prompt, run 1:
<https://github.com/harvest7777/codex-parallel-graphviz-run/compare/main...agent-work>
- Codex, parallelism prompt, run 2:
<https://github.com/harvest7777/codex-parallel-2-graphviz-run/compare/main...agent-work>

Source. GraphViz at commit 23b06521ab916e7f4f2ae2f9ef8413a8ccc66601, from <https://gitlab.com/graphviz/graphviz.git>. Both agent branches are named agent-work and branch from that commit.

Build.

```
cmake -S <sandbox> -B <build> \
  -DCMAKE_INSTALL_PREFIX=<prefix> \
  -DWITH_POPPLER=OFF -DWITH_GVEDIT=OFF \
  -DCMAKE_BUILD_TYPE=Release
cmake --build <build> -j12 && cmake --install <build>
```

Identical for all nine sandboxes. On macOS a Bison ≥ 3.0 path must be passed explicitly, as the system Bison is too old for the GraphViz build.

Measurement. hyperfine 1.20.0, 10 warmup and 10 timed runs, all nine binaries for a given graph in a single invocation, repeated over four command orderings with a different binary leading each time, at each of two node counts (SIZE=5000 and SIZE=10000). Graphs are produced by `graphs/gen-graphs.py`, which writes both sizes from one fixed seed. Raw per-run timings for every configuration are in `results/` as hyperfine JSON exports, alongside the per-run phase logs in `results/phases/`; result filenames carry the graph name, so the two sizes sit side by side.

Figures. Every figure is regenerated from the raw JSON by a single script, `new paper/figures/make_figures.py`.

Environment. Apple M4 Pro, 12 cores (8 performance, 4 efficiency), 24 GB unified memory, macOS 26.5.2, Apple Clang 17.0.0.

Phase instrumentation. The patch that reports per-phase timings is a single commit on `lib/dotgen/dotinit.c`, applied by cherry-pick to all nine sandboxes so that the nine diffs are byte-identical. It is inactive unless `GV_PHASE_TIMING` is set in the environment.

Agents. `claude-opus-5` (Claude Code) and `gpt-5.6-terra` (Codex), each invoked four times autonomously — two runs under each of the two prompts, both in `prompts/`. No run had network access to the evaluation harness or benchmark graphs, or to any other run’s sandbox. Session dumps giving each run’s time and token usage are in `results/dumps/`, one file per run, named for the sandbox it came from.

7 Conclusion

Eight agent runs, given identical checkouts of GraphViz and prompts naming no target, all located the same expensive code. Finding the bottleneck was evidently not the difficult part of the task.

What separated them was breadth. Every Claude run carried on into network simplex, which serves two further phases, and improves three of five on every shape and size we tested — $1.39\times$ to $4.02\times$ end-to-end. Codex’s one shipping run optimized the phase it identified first and measures $1.02\times$ to $1.33\times$; on the sparsest shape that optimization works, at $1.32\times$ on its phase, and still yields nothing end-to-end because the phase is 12.4% of the workload.

Codex’s other three runs shipped nothing, each after building a change, measuring it, and judging it not worth keeping. We take that at face value. It does mean a two-run study would have described whichever run it happened to draw, which is the strongest argument in this paper for repeating agent runs before characterizing an agent.

Asking for concurrency changed what the agents built but not uniformly what it bought. Two Claude runs under that prompt land close together end-to-end while only one of them owes its result to threads, and the same prompt moved Codex to prototype concurrency three times and ship it zero times.

The ceiling an optimization can reach is a property of the input, not of the program. At fixed node count, crossing minimization ranges from 12.4% to 64.8% of layout time across our three shapes, and on the sparsest one every Claude run exceeds what any crossing-minimization optimization could achieve. Doubling the node count lowers every ceiling — to 1.08, 1.42 and $2.04\times$ — because that phase’s share shrinks as graphs grow. A single benchmark graph cannot show the first effect, and a single graph size cannot show the second.

The results are modest and we have tried to report them at the size they are. The protocol may be the more durable contribution: balance command ordering and take the floor from the baseline’s own spread across those orderings; keep the evaluation harness out of the agents’ reach; forbid the shortcuts that manufacture speedups; measure phases directly rather than by subtraction; re-measure threaded binaries at one thread before calling a result parallel. Each caught something a simpler design would have reported as a finding.

Immediate future work is to extend to a third graph size and to non-synthetic inputs, to run the concurrent code under a dynamic race detector on a platform where one works, and to complete the correctness checks named in Section 6.

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